

Exponential Expressions

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Lesson 3 — Exponential Expressions	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.NSO.1.1

2. Standards

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Must. Two or three laws in one expression, whole-number exponents and rational bases; generate an equivalent expression and evaluate it.

Must not. No negative exponents yet. The laws never cross addition or subtraction, and order of operations stays at six steps or fewer.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II and board 9 — the same expression is carried to the end by two different plans, written side by side, and the two answers are compared.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1 through 4 — four multi-law expressions, one each, written without a reference sheet.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Board 5 — a sixth power over 2^{10} lands on 4; students say why an answer that small is reasonable before they trust it.

4. Learning Target and Essential Question

Learning target. I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

Essential question. How can the laws of exponents be used to generate equivalent numeric expressions?

Building on. Lessons 3.01–3.02 — the seven laws; MA.7.NSO.2.1 — order of operations with rational numbers.

Working toward. MA.8.NSO.1.3 — the same combined work with integer exponents (3.06–3.07); MA.8.AR.1.1 — with variable bases (Thread A).

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the 'today' line and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	14	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (14 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Winston works inside the parentheses first and Yvette expands first. Why is the order yours here, but not in $2 + 3 \cdot 4$?	DOK 3
$3 \cdot 2^0$ sits inside a sixth power and changes nothing. What would have to change about that factor for it to matter?	DOK 3
Which plan for $(0.5^3)^2 \div 0.5^2$ uses fewer digits, and why?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A warehouse holds 6^4 boxes in 6^2 equal rows. Each box weighs 3 kilograms. How many boxes are in each row?	36	1296 — the 6^2 rows were ignored [the quotient was never taken]; 6^6 or 46,656 — the exponents were added [confused with product of powers]; anything built from 3 — the unneeded figure was used
2	$(3^0 \cdot 3^2)^3$	729	243 — added the exponents $2 + 3$ in the power of a power step [confused with product of powers]; 0 — zero exponent read as 0
3	$\frac{(0.5^3)^2}{(0.5)^2}$	0.0625	0.125 — divided the exponents $6 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]; 0.015625 — 0.5^6 , the denominator ignored
4	$\left(\frac{2.5^5}{2.5^3}\right)^1$	6.25	2.5 — the outside 1 was treated as the final exponent [identity law misapplied]; 15.625 — used 2.5^3
5	$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$ Which is the value?	D — 4	(A) $(2^2)^6$ written as 2^8 by adding $2 + 6$, then $2^8 \div 2^{10}$ [power of a power confused with product of powers]; (B) added the exponents $12 + 10$ instead of subtracting [confused with product of powers]; (C) $3 \cdot 2^0$ read as 0, so the whole numerator is 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
6	$\frac{0.2^3}{2^3} \cdot 0.1^2$ Which expression is equivalent?	C — 0.1⁵	(A) quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]; (B) the 2^3 in the denominator was ignored [order of operations: the fraction bar is a grouping]; (D) multiplied the exponents $3 \cdot 2$ [confused with power of a power]
7	$0.1^2 \cdot (0.1^1)^2$	0.0001	0.001 — 0.1^3 , the outer square on $(0.1^1)^2$ dropped, leaving $0.1^2 \cdot 0.1^1$ [power of a power step skipped]; 0.01 — stopped at 0.1^2
8	$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$	-64	64 — dropped the sign; 'undefined' — $(-4)^0$ in the denominator read as 0, so a division by 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
9	Winston writes $(2^6/2^4)^3 = (2^2)^3$. Yvette writes $(2^6/2^4)^3 = 2^{18}/2^{12}$. Do they get the same value? Explain.	Yes — both give $2^6 = 64$	'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. The seven laws from Notes I stay on the desk as a half-sheet. Before writing, the student says the plan out loud in two words — 'inside first' or 'expand first' — and the teacher confirms the plan, not the answer.

ELL. Plan and evaluate are the two verbs on every slide today. Write PLAN = decide the order and EVALUATE = find the number on the board, and ask for the plan in words before any board goes up.

Enrichment. Ask for a single power of 2 equal to $(2^3 \cdot 4^2) \div 8$, then for one expression that two different laws can both finish, to the same value.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to carry both plans to the end and show they land on the same value. Read the boards, not the papers.

Vocabulary. **equivalent expressions; evaluate; plan**

Materials. Whiteboards and markers. Calculators are allowed on everything; the decimal values in the bank (5.0625, 2,293.45007, 9,605.9601) are calculator work once the single power is found.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate expressions using properties of exponents — ZE9.